CCR root matching on all 21 candidates

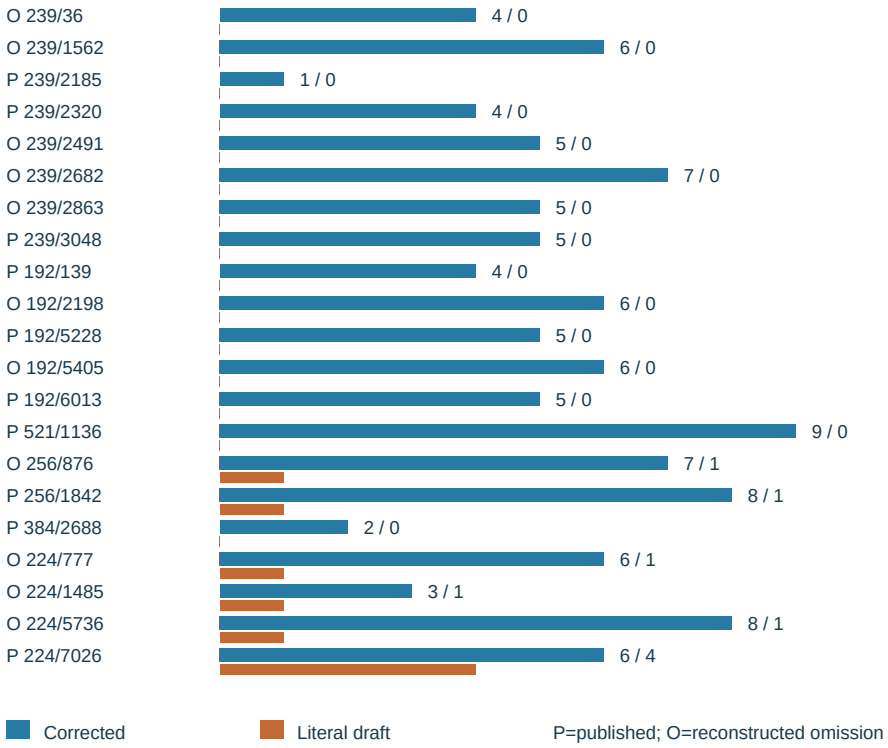
2026-10-27 | T-5518 | Observation report; no formal hypothesis decision.

The historical selector remains unknown. Corrected recovery finds all 112 available cyclic kernels across 147 curve/degree cells. All 11 omissions and all 10 publications enter a favorable case at degree 7 or higher. No cohort projection collision is found.

1. Two conversions, every root pairing

Degrees: 3,5,7,11,13,17,19. The observer receives p,A,B,ell,index; labels and known orders enter independent post-audit only. Fresh mathematical projection root sets replace historical q-series tables. This is a bounded reconstruction, not the original executable.

Distinct cyclic kernels recovered: corrected / literal conversion



Unit: distinct kernels summed over seven degrees. n=21 curves,147 cells; exact counts, no error bars.

CCR-ROOT-MATCHING-01;848 cohort pair reconstructions; all nine literal kernels occur at degree 3.

Each arm tries 424 cohort root tuples. Corrected conversion has 172 successful factor pairings but112 distinct cyclic kernels. Literal conversion has 69 factor pairings but only nine distinct kernels, all at degree 3. There are no literal successes at degree 5 or higher.

At degree 3 the kernel depends on S1 alone; different quotient-coefficient pairs can produce the same kernel. Counting those as independent successful kernels would inflate the result. The chart plots distinct kernels per curve/degree, summed over the declared levels.

2. Source predicates and independent checks

CCR page 6 advances to the next auxiliary prime when no base-field S1 root exists, then selects a non-repeated root. Page 18 tries pairings of independently found Anew and Bnew roots and tests the putative division factor. It does not prescribe rejecting a curve after one bad guess.

Independently verified: full division polynomials and scalar x-coordinates; projected-polynomial annihilation plus Krylov-rank minimality; complete rational root sets; both power-sum conversions; every Newton kernel and division remainder; cyclic closure; sign-ambiguous Frobenius trace sets containing known traces.

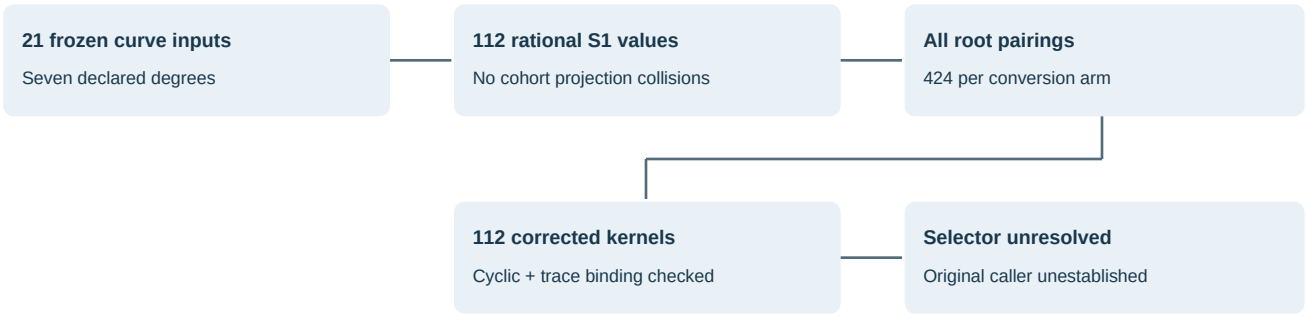
Including controls: 456 projection-minimality proofs,735 scalar-coordinate checks,932 pair reconstructions and 249 cyclic-pair trace checks. Cohort projection degrees are always ell+1, so none has the tested coordinate collisions or repeated S1 values. All 112 rational S1 values recover a corrected cyclic kernel.

Controls expose the condition that is absent from the cohort

F_101 model	Projection degrees at 7	Observed result
A=1,B=1 (ordinary)	8,8,8	Published S1=31,Anew=19,Bnew=26 kernel reproduced
A=0,B=1 (j0)	7,7,4	S1 fibers 6,3,3: repeated value rejected by source root rule; other two recover
A=1,B=0 (j1728)	8,4,8	Projection collision; no rational roots at 7

The j0 control has one unfiltered S1 value without a cyclic kernel. The paper already excludes that repeated root. This demonstrates the coordinate complication; it is not evidence that the documented method rejects the control curve. No cohort case has it.

A favorable degree 19 control exercises the highest recurrence index before cohort calls. The published-example A/A captures match exactly after normalizing only index. Direct control orders 105,102,100 agree with local LMFDB isogeny classes 1.101.d,1.101.a,1.101.ac; those rows do not identify individual models.



3. Complete custody and remaining limits

Attempt 01 stops at its required collision-control gate, before any cohort call. Its 32-case generic scan found no collision. Attempt 02 starts147 cohort cells;101 complete and 46 remain partial because GP factorial() yields a real value. The process still exits 0 and prints its final marker: stderr and per-cell framing are essential.

Attempt 03 uses exact integer factorials and the maximum-degree control. All 147 cohort cells and five controls finish. Every unsuccessful capture and amendment remains in the export; operational failures are not curve rejections. A pre-run derivative-syntax correction started no observer call.

Three external GP observer processes are recorded. Across retained attempts: 294 cohort cells started,248 completed,46 partial. Final native diagnostic wall/user CPU/system CPU: 4.730161/4.643442/0.022960 seconds. Independent audit wall: 80.286795 seconds; its CPU time was not captured. These modern ARM64 observations are not performance or historical-runtime comparisons.

Raw evidence and replay

Archive: 65 sealed files; 18,671,614 bytes.
SHA-256: 4f4c4478b4967089685dd626625451afce409d508c749e435cbdb970be32ea03

After extraction, use python-flint 0.9.0 and run:
`python3 -B evidence/verify.py`
The full audit passed after extraction into a different directory. No external producer or network is invoked. FLINT runs in process. Exact fields/models/orders/seeds/UIDs are in comparison.csv and the sealed cohort.

Local LMFDB manifests were complete before querying. The av_fq source was postgresql://devmirror.lmfdb.xyz:5432/lmfdb/public.av_fq_isog, extraction ended 2026-10-19T16:45:49.243703+00:00. CONTROL-COUNTS.json binds exact models, direct counts, labels and provenance. No challenge crosswalk is claimed.

What remains unestablished

Original q-series builder and tables; executable lineage; historical root enumeration; larger auxiliary degrees; full sign recovery, CRT/resource policy and candidate visitation. The documented root and pairing rules tested here do not separate the labels. No transfer or endomorphism selection rule is identified.

Primary sources: CCR draft pages 6,8-9,18 in Morain's author archive; Morain 1995, printed 274-275, doi:10.5802/jtnb.143. Full URLs and digests are in report.md and source receipts. The CCR scan version remains uncertain. Publisher PDFs/images and interpreter binaries are excluded explicitly.

Canonical store push was refused at HeadObject403. Git retains the exact archive under the existing PR exception. Inherited immutable A/A-record collisions remain; no green merge-readiness claim or formal hypothesis change is made.